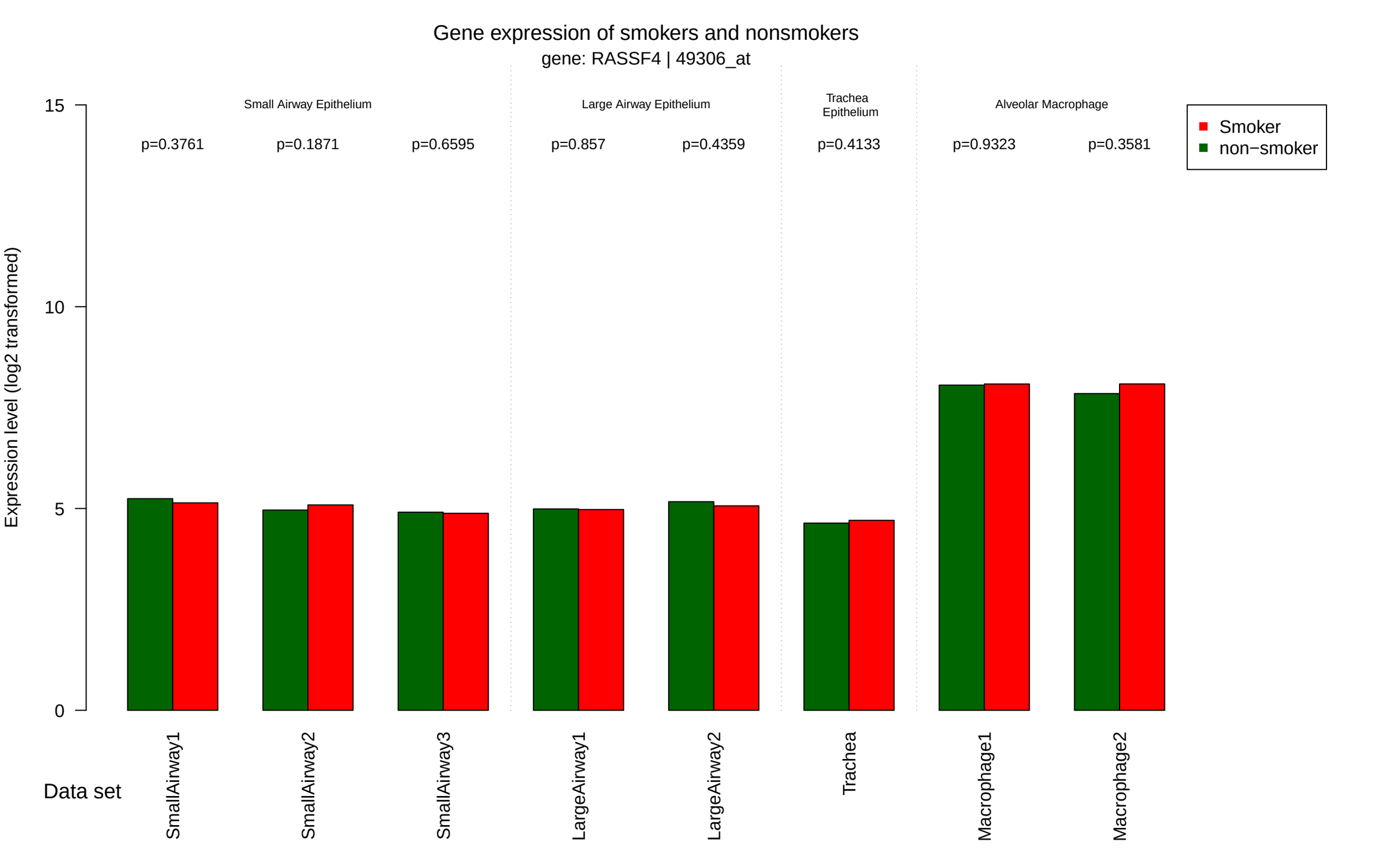
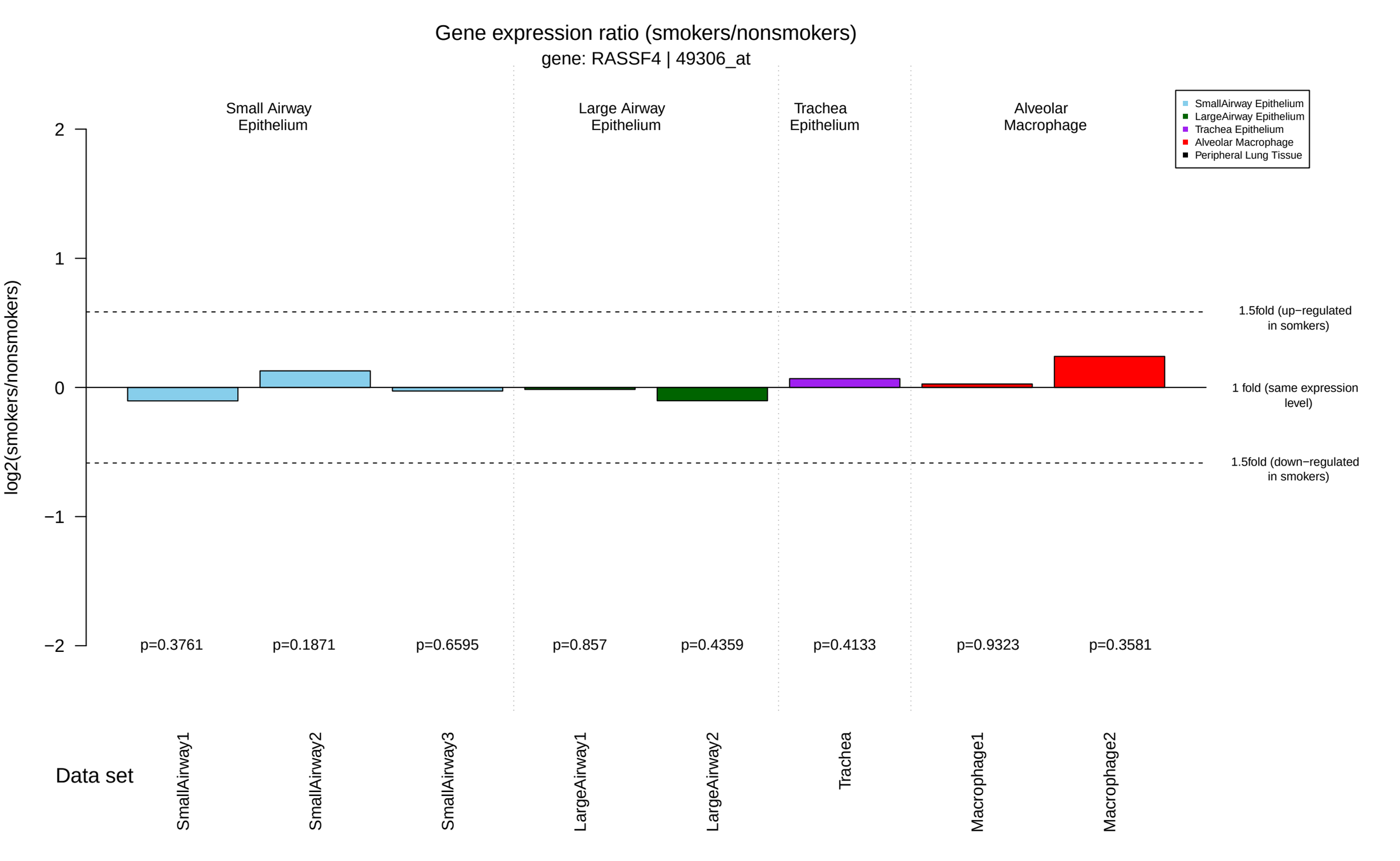
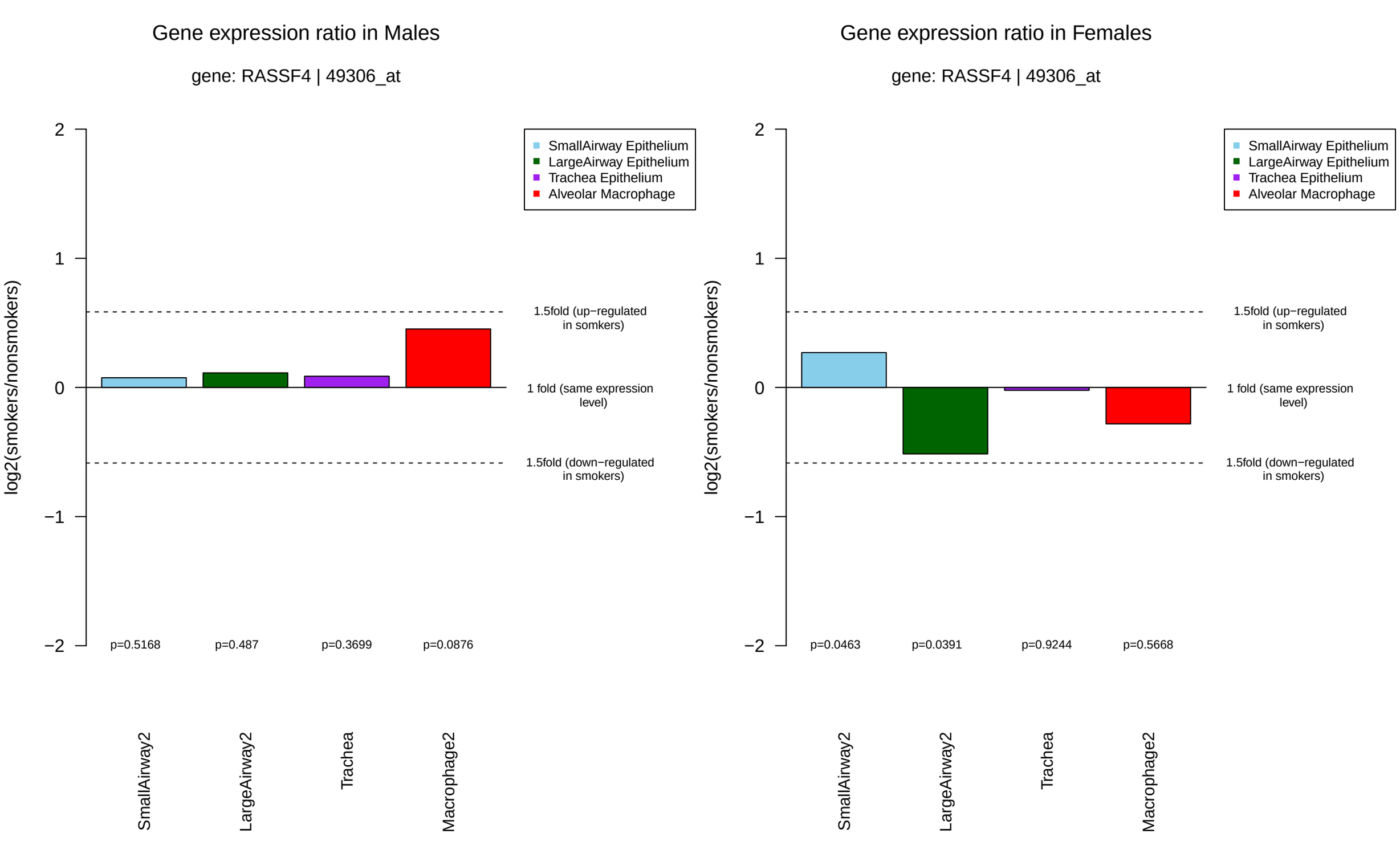
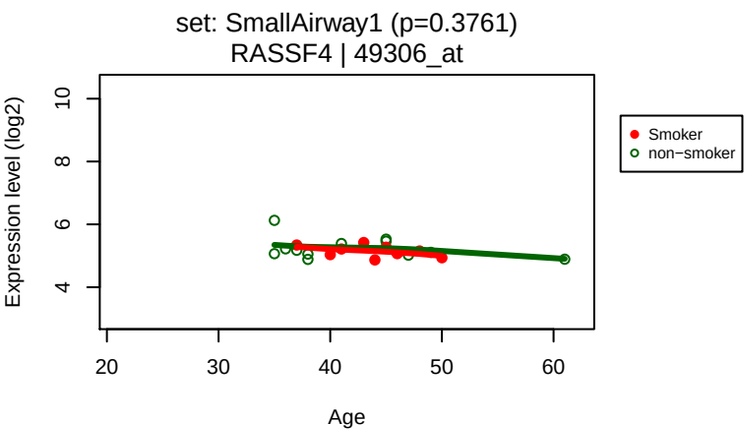
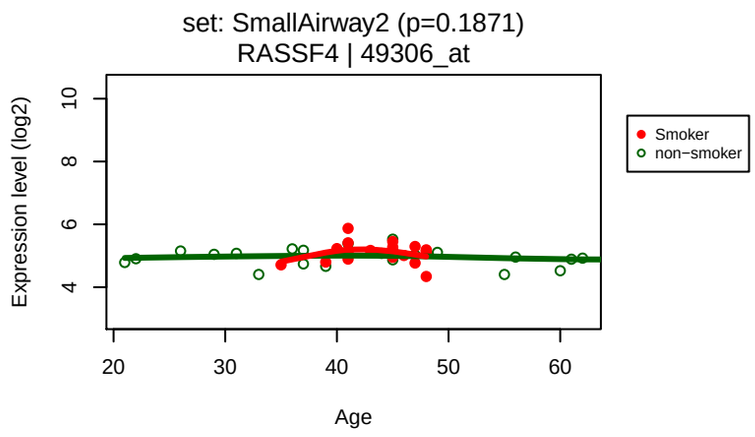
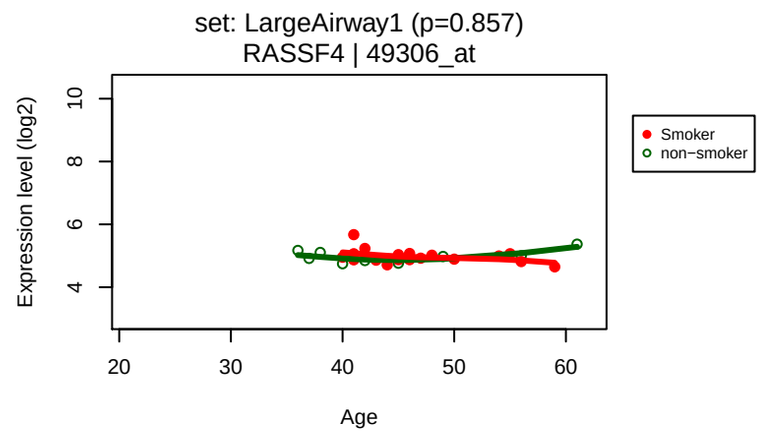

Gene expression of smokers and nonsmokers
gene: RASSF4 | 49306_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3761
p=0.1871
p=0.6595
p=0.857
p=0.4359
p=0.4133
p=0.9323
p=0.3581
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

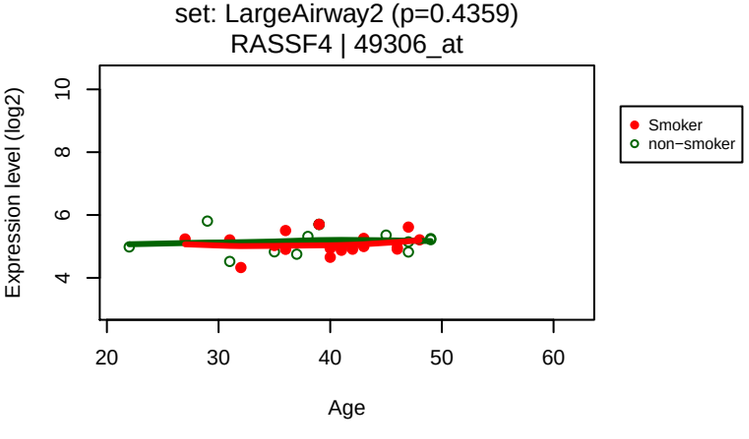

Gene expression ratio (smokers/nonsmokers)
gene: RASSF4 | 49306_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3761
p=0.1871
p=0.6595
p=0.857
p=0.4359
p=0.4133
p=0.9323
p=0.3581
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

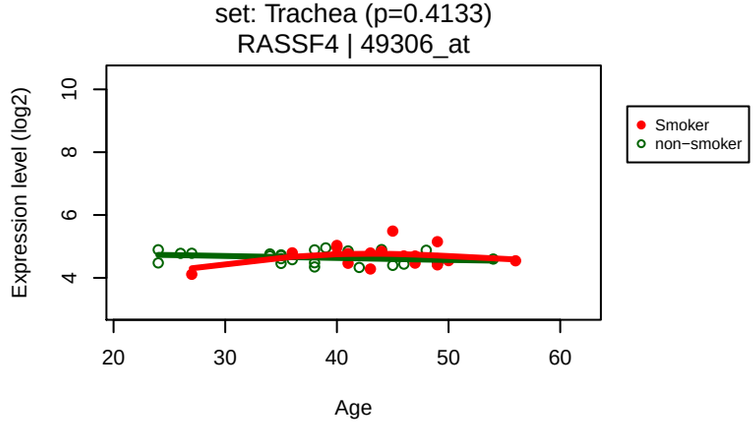

Gene expression ratio in Males
gene: RASSF4 | 49306_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5168
p=0.487
p=0.3699
p=0.0876
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASSF4 | 49306_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0463
p=0.0391
p=0.9244
p=0.5668
SmallAirway2
LargeAirway2
Trachea
Macrophage2

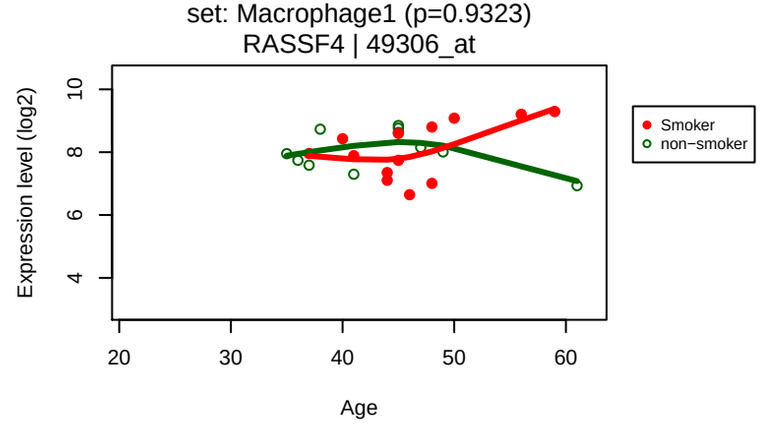

set: SmallAirway1 (p=0.3761)
RASSF4 | 49306_at
Expression level (log2)
Age
Smoker
non−smoker

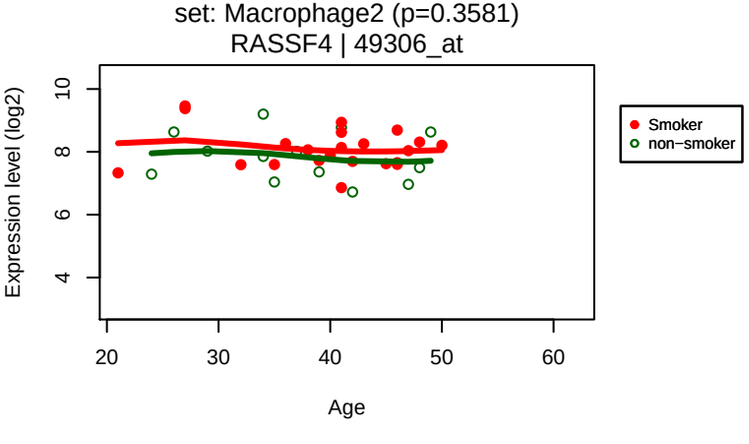

set: SmallAirway2 (p=0.1871)
RASSF4 | 49306_at
Expression level (log2)
Age
Smoker
non−smoker

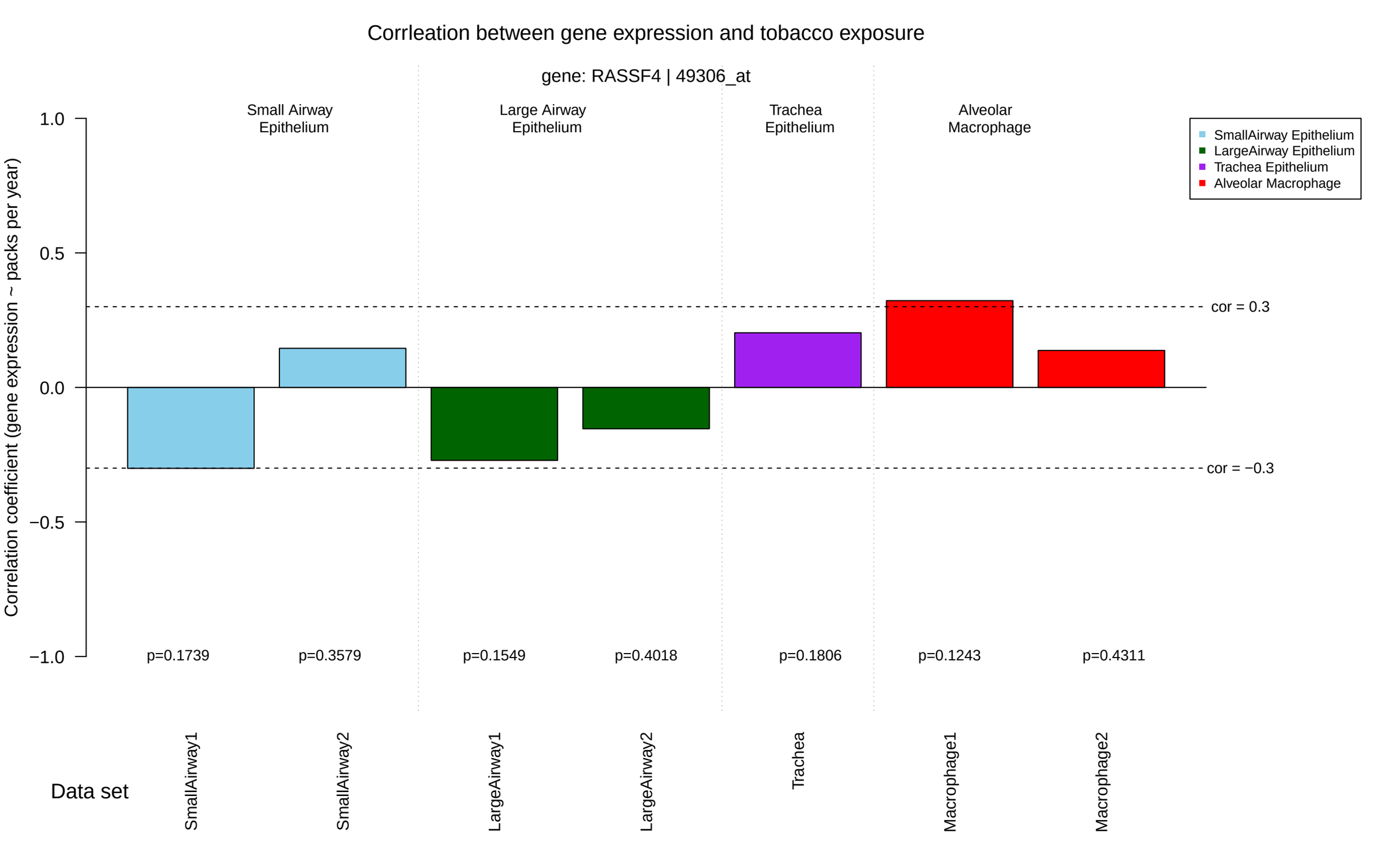

set: LargeAirway1 (p=0.857)
RASSF4 | 49306_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4359)
RASSF4 | 49306_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4133)
RASSF4 | 49306_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9323)
RASSF4 | 49306_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3581)
RASSF4 | 49306_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASSF4 | 49306_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1739
p=0.3579
p=0.1549
p=0.4018
p=0.1806
p=0.1243
p=0.4311
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2